

9 Counting And Probability

9.1 Introduction

9.1.1 Sample Space

A **sample space** is the set of all possible outcomes of a random process or experiment. An **event** is a subset of a sample space.

9.1.2 Equally Likely Probability Formula

If S is a finite sample space in which all outcomes are equally likely and E is an event in S , then the **probability of E**, denoted $\mathbf{P(E)}$, is

$$P(E) = \frac{\text{the number of outcomes in } E}{\text{the total number of outcomes in } S} \quad (1)$$

The notation is as follows: For any finite set A , $N(A)$ denotes the number of elements in A .

9.1.3 The Number of Elements in a List

If m and n are integers and $m \leq n$, then there are $n - m + 1$ integers from m to n inclusive.

9.2 Possibility Trees and the Multiplication Rule

9.2.1 The Multiplication Rule

If an operation consists of k steps and

the first step can be performed n_1 ways, the second step can be performed in n_2 ways [*regardless of how the first step was performed*]; the k th step can be performed in n_k ways [*regardless of how the preceding steps were performed*],

then the entire operation can be performed in $n_1 n_2 \cdots n_k$ ways.

9.2.2 Number of Permutations

For any integer n with $n \geq 1$, the number of permutations of a set with n elements is $n!$.

9.2.3 R-Permutation

An **r-permutation** of a set of n elements is an ordered selection of r elements taken from the set of n elements. The number of r-permutations of a set of n elements is denoted $\mathbf{P(n, r)}$.

If n and r are integers and $1 \leq r \leq n$, then the number of r -permutations of a set of n elements is given by the formula

$$P(n, r) = n(n-1)(n-2) \cdots (n-r+1)$$

or, equivalently,

$$P(n, r) = \frac{n!}{(n-r)!}$$